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CENTRE DE SYNTHÈSE ET D'ANALYSE SUR LA BIODIVERSITÉ

INTRODUCTION AUX MÉTA-ANALYSES ET REVUES SYSTÉMATIQUES

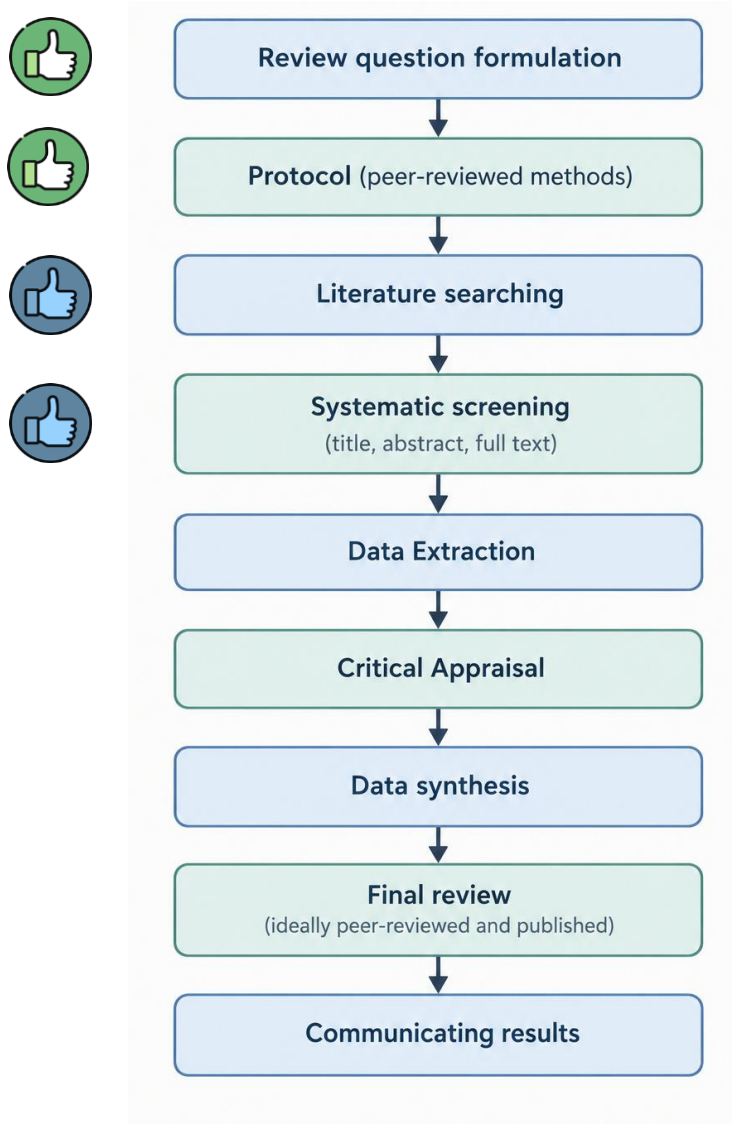
RepOrting Standards for Systematic Evidence Syntheses (ROSES)

Joseph LANGRIDGE – Chargé de mission scientifique « Revues systématiques et synthèses »

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Recap: steps taken



ROSES

What is it?

ROSES = RepOrting standards for Systematic Evidence Syntheses in Environmental management

What is provided?

- **Flow diagrams**: created by a team of researchers (CEE) experience in conducting systematic reviews in the environmental field, inspired by pre-existing tools from other disciplines (such as PRISMA in the medical field).

- **Forms**: specifically designed for systematic reviews and systematic maps in the field of environmental management developed by the Collaboration for Environmental Evidence (CEE).

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Transparency, rigor, and traceability

key objectives

Without transparent reporting, even well-designed reviews will fail to show their methodological strengths, undermining their utility in decision-making contexts **(Pussegoda et al. 2017)**

A ***persistent problem*** in Evidence synthesis:

- In “traditional”/ “non-systematic” reviews, the details of the **different steps and volumes** are only very rarely presented; they may sometimes be inferred, but are most often **completely omitted**. Decisions are not documented.

- The review is therefore *not reproducible*!

- The reader cannot understand how the final result was obtained (lack of primary studies? high rate of inaccessible PDFs, etc.).

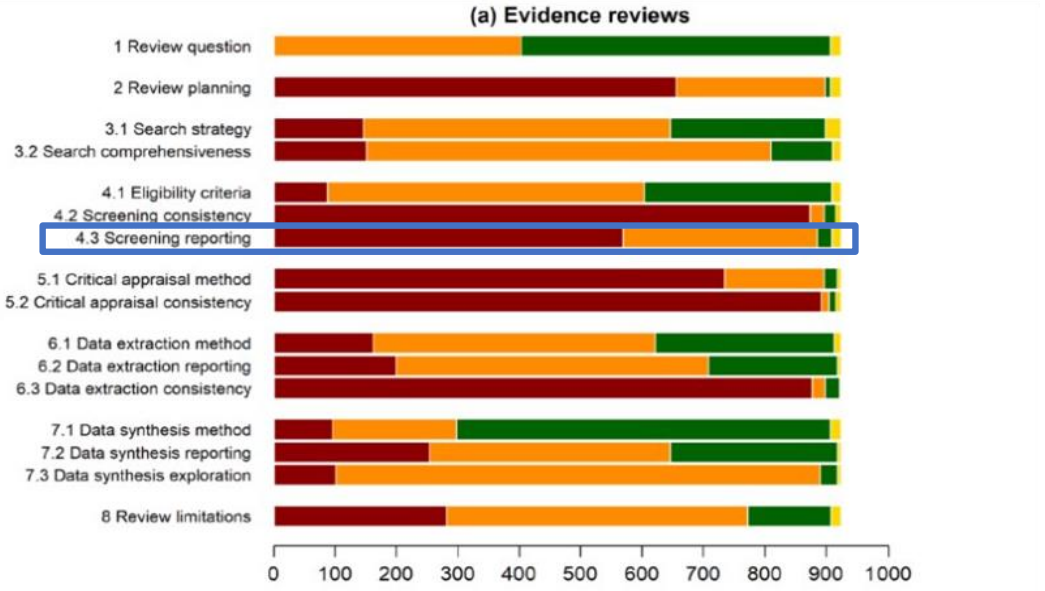
Source: Pussegoda *et al.* 2017.*Syst Rev*

Standards and reporting

COMMENTARY Open Access

Standards of conduct and reporting in evidence syntheses that could inform environmental policy and management decisions

Andrew S. Pullin[1,11*], Samantha H. Cheng[2], Josephine D'Urban Jackson[3], Jacqualyn Eales[4], Ida Envall[5], Salamatu J. Fada[6,7], Geoff K. Frampton[8], Meagan Harper[9], Andrew N. Kadykalo[9], Christian Kohl[10], Ko Konno[11], Barbara Livoreil[12], Dakis-Yaoba Ouédraogo[13], Bethan C. O'Leary[14,15], George Pullin[16], Nicola Randall[17], Rebecca Rees[18], Adrienne Smith[19], Romain Sordello[20], Eleanor J. Sterling[21], Will M. Twardek[22] and Paul Woodcock[23]



Source: Pullin *et al.* 2022. *Environ Evid*

Standards and reporting

CEE Synthesis Assessment Tool: **CEESAT**



The CEESAT workshop group. From left: Jacqui Eales, Neal Haddaway, Ruth Garside, Nicola Randall, Barbara Livoreil, Andrew Pullin, Geoff Frampton, Christian Kohl and Biljana Macura.



1. THE REVIEW QUESTION

Rationale: A well-defined question (or hypothesis) is crucial for assessing the reliability of subsequent decisions on searching and screening for eligible studies, as well as forming the basis for critical appraisal of study conduct and for data extraction and synthesis.

1.1 Are the elements of the review question clear?

Tick whichever statement applies

- ○ Gold: The review question or hypothesis is clearly stated and clearly defines key elements, (e.g. PICO, PECO, PO, PIT, etc.) correctly, such as the subject or population of interest, the intervention or exposure type, the comparator and valid measures of outcome.
- ○ Green: The review question or hypothesis is clearly stated, and key-elements are mentioned although not formally defined in terms of PICO, PECO, PO, PIT, etc.
- ○ Amber: The question or hypothesis is stated in broad terms but key-elements are unclear or poorly defined.
  OR
  Question or hypothesis not stated but problem or issue is stated such that a question can be inferred
- ○ Red: A question, hypothesis or problem is not stated
  OR
  There is no stated objective to provide an answer to a question or test of a hypothesis.
  OR
  The article does not contain an evidence synthesis (e.g. primary research or descriptive overview)

CEE Synthesis Assessment Tool

CEESAT

2.1. Is there an a-priori method/protocol document?

- ○ Gold: The review cites a separate a-priori protocol or documented pre-defined method containing details of proposed conduct of all review and synthesis stages (e.g. question, search, eligibility screening, critical appraisal, data extraction and synthesis)
  NOTE: If the protocol is not specifically written for the review then the assessment should be green rather than gold.
  AND
  It is linked from the synthesis (e.g. as supplementary material or hosted on a separate website)
  AND
  It was publicly accessible prior to the conduct of the review
  AND
  It was submitted to an independent body for peer review and publication.
- ○ Green: The review cites a separate accessible a-priori protocol or documented pre-defined method containing details of conduct of all review and synthesis stages (e.g. question, search, screening, critical appraisal, data extraction and synthesis)
  AND
  It is linked from the synthesis (e.g. as supplementary material or hosted on a separate website)
  AND
  It was publicly accessible prior to the conduct of the review.
- ○ Amber: The review cites a separate accessible a-priori protocol or documented pre-defined method, but this does not contain all details of conduct of all review and synthesis stages or was not publicly accessible prior to the conduct of the review
  OR
  The review includes a defined methods section (not a-priori) listing the synthesis stages conducted and providing sufficient detail to enable the method to be replicated (therefore this standard is met only if all of criteria 3.1, 4.1, 6.1 & 7.1 are rated green or above).
- ○ Red: There is no protocol and the review methods are not clearly defined in the methods section of the review or there are no methods reported.

4.3. Are eligibility decisions transparently reported?

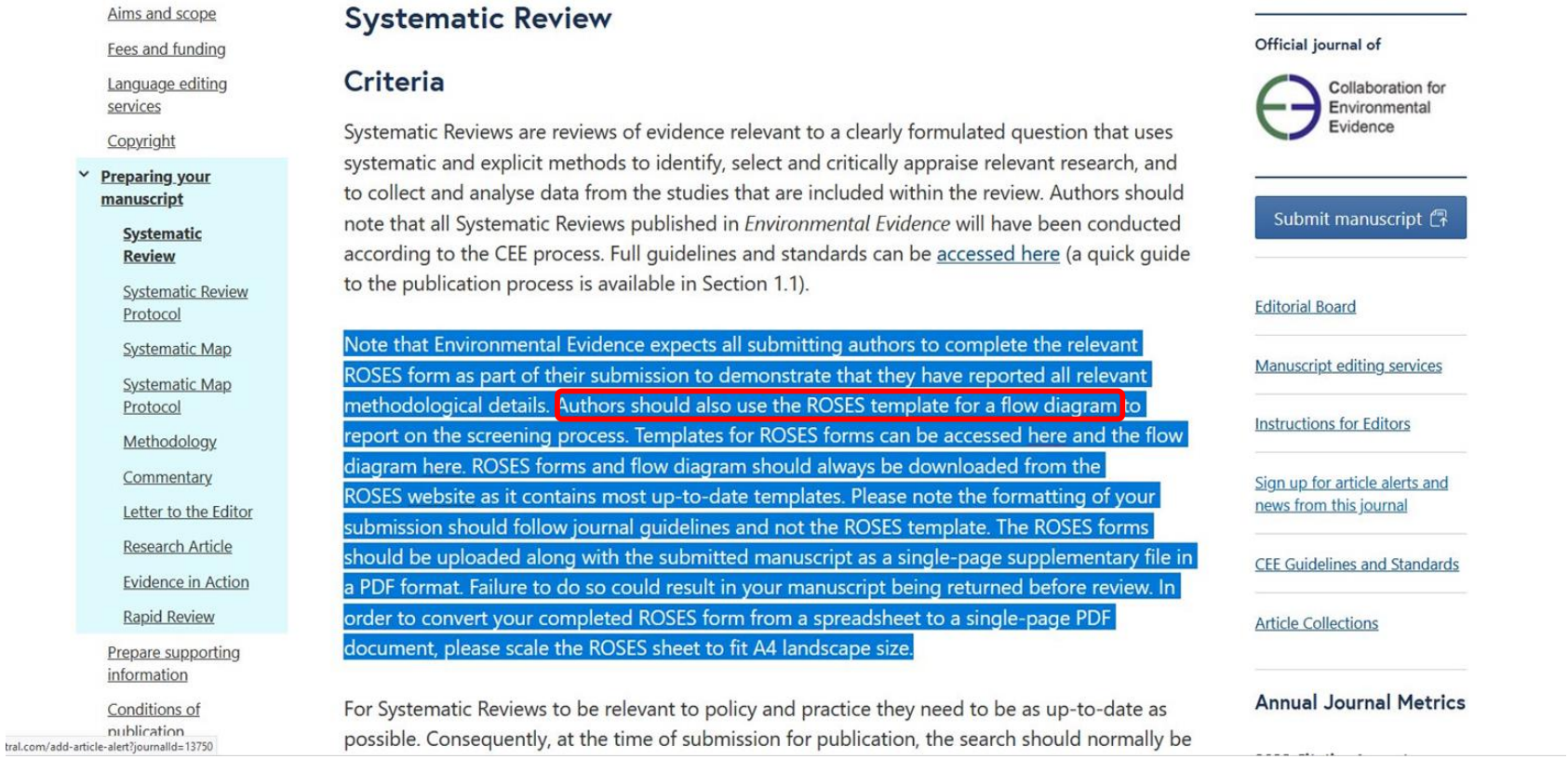
Rationale: Listing all articles that were screened for eligibility and indicating whether each was included or excluded in data synthesis (usually as supplementary material), makes it clear whether potentially relevant studies have been omitted according to the eligibility criteria or were not captured by the search. Documenting the reasons for article exclusion at full text is essential for transparency.

- ○ Gold: The total number of articles and number of unique articles found during the searches (after removal of duplicates) is presented
  AND
  The number excluded at each stage of the screening process is fully presented (e.g. in a flow diagram or table)
  AND
  Reasons for exclusion of each article/study considered at full-text are presented (e.g. in an appendix)
  AND
  A list of eligible (included) articles/studies is presented as a separate list or in tables (not just included in reference list).
- ○ Green: The total number of articles found by the search and numbers excluded at each stage of the screening process is reported but some other aspects missing (e.g. number of unique articles or articles unobtainable)
  AND
  Reasons for exclusion of each article/study considered at full-text are presented (e.g. in an appendix)
  AND
  A list of eligible (included) articles/studies is presented as a separate list or in tables (not just included in reference list).
- ○ Amber: The total number of articles found by the search and numbers excluded during the screening process is reported (or inferable) but some other aspects missing (e.g. number of unique articles or articles unobtainable or reasons for exclusion at full text)
  AND
  A list of eligible (included) articles/studies is presented as a separate list or in tables (not just included in reference list).
- ○ Red: No to either or both of the amber criteria above

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Standards and reporting

Requirements



Aims and scope
Fees and funding
Language editing services
Copyright
Preparing your manuscript
Systematic Review
Systematic Review Protocol
Systematic Map
Systematic Map Protocol
Methodology
Commentary
Letter to the Editor
Research Article
Evidence in Action
Rapid Review
Prepare supporting information
Conditions of publication

tral.com/add-article-alert?journalId=13750

Systematic Review

Criteria

Systematic Reviews are reviews of evidence relevant to a clearly formulated question that uses systematic and explicit methods to identify, select and critically appraise relevant research, and to collect and analyse data from the studies that are included within the review. Authors should note that all Systematic Reviews published in *Environmental Evidence* will have been conducted according to the CEE process. Full guidelines and standards can be accessed here (a quick guide to the publication process is available in Section 1.1).

Note that Environmental Evidence expects all submitting authors to complete the relevant ROSES form as part of their submission to demonstrate that they have reported all relevant methodological details. Authors should also use the ROSES template for a flow diagram to report on the screening process. Templates for ROSES forms can be accessed here and the flow diagram here. ROSES forms and flow diagram should always be downloaded from the ROSES website as it contains most up-to-date templates. Please note the formatting of your submission should follow journal guidelines and not the ROSES template. The ROSES forms should be uploaded along with the submitted manuscript as a single-page supplementary file in a PDF format. Failure to do so could result in your manuscript being returned before review. In order to convert your completed ROSES form from a spreadsheet to a single-page PDF document, please scale the ROSES sheet to fit A4 landscape size.

For Systematic Reviews to be relevant to policy and practice they need to be as up-to-date as possible. Consequently, at the time of submission for publication, the search should normally be

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ROSES

flow diagrams

ROSES flow diagrams

What is it?

- It is a **flowchart** that allows readers to quickly visualize the **entire systematic map or review process** and the reduction in study/article volumes through the different stages.

- It should be included as a **figure** in the manuscript of a systematic map/review.

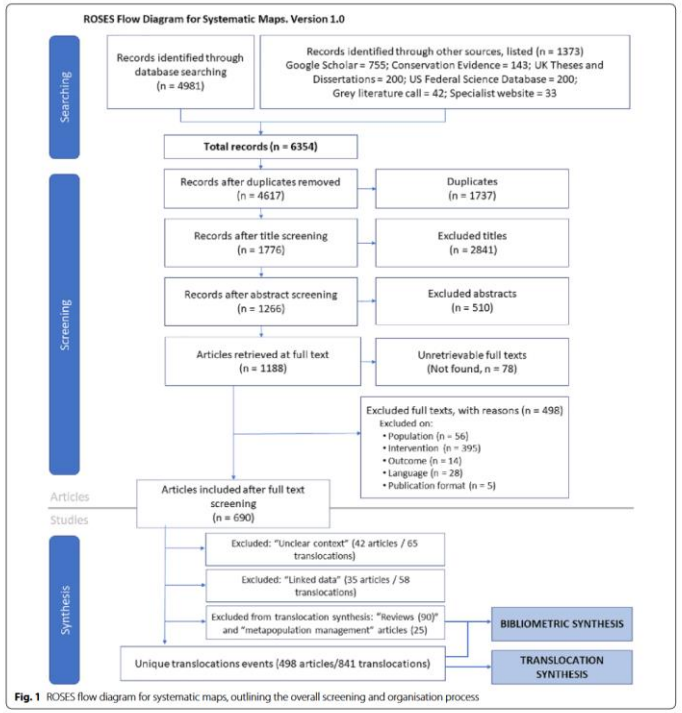
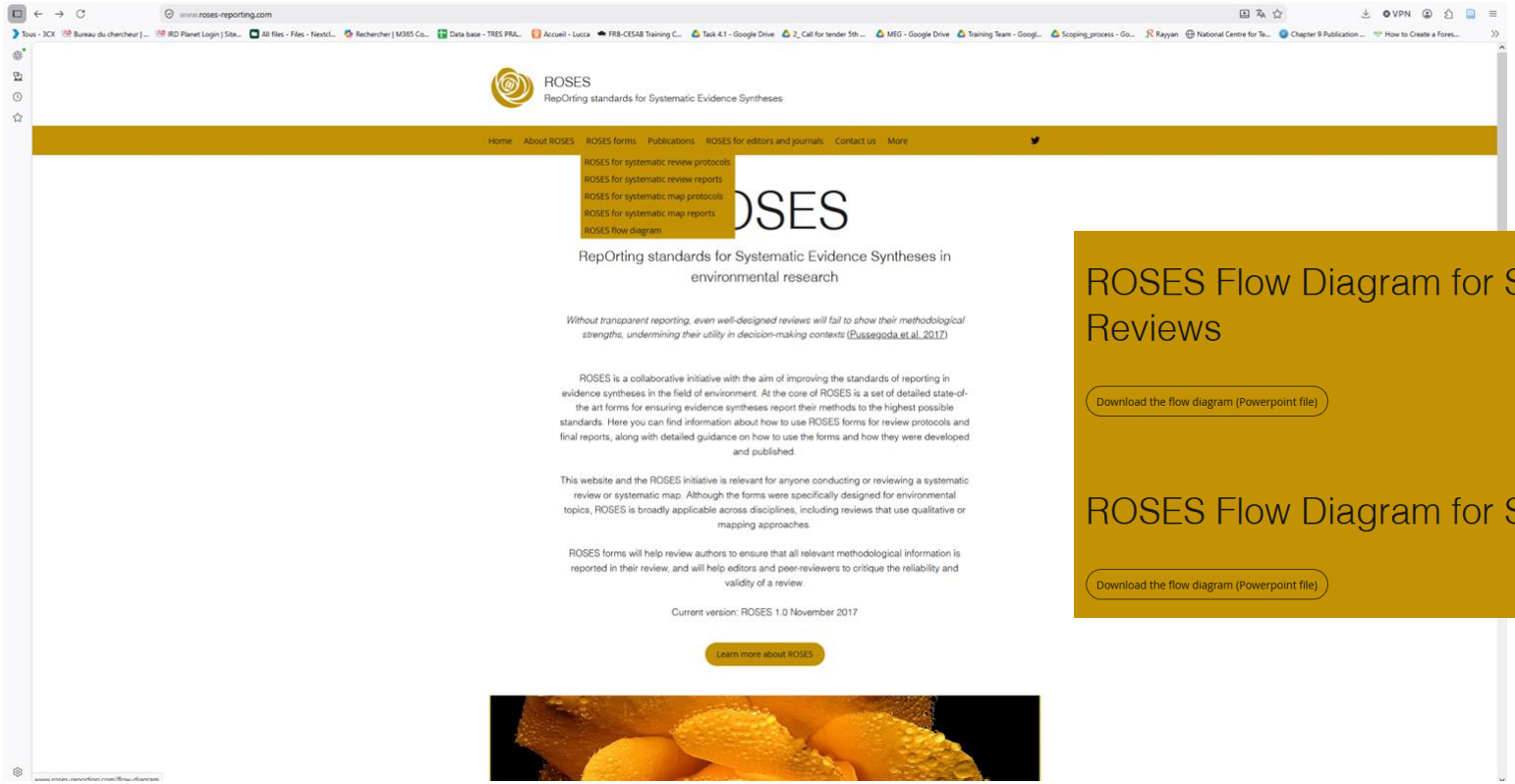


Fig. 1 ROSES flow diagram for systematic maps, outlining the overall screening and organisation process

programmes. Human-wildlife rescue operations accounted for 33 translocations, unsurprisingly only animals are concerned, of which 17 transfers did not document if individuals were released into resident populations or not (Table 3). Regarding rewilding, which was considered in the context of reintroducing a species to restore lost ecological functions, equated to only 8 translocations (4 in Argentina, 2 in Brazil and 2 in Germany), and in each case the relocation of a mammal. One study, in Brazil, aimed at reintroducing the red-humped agouti and the brown howler monkey, in Tijuca National Park, Rio de Janeiro; two species that are important seed dispersers, which may be particularly effective at restoring such ecological processes [64]. Finally, 64 translocations had completely

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ROSES flow diagrams



ROSES Flow Diagram for Systematic Reviews

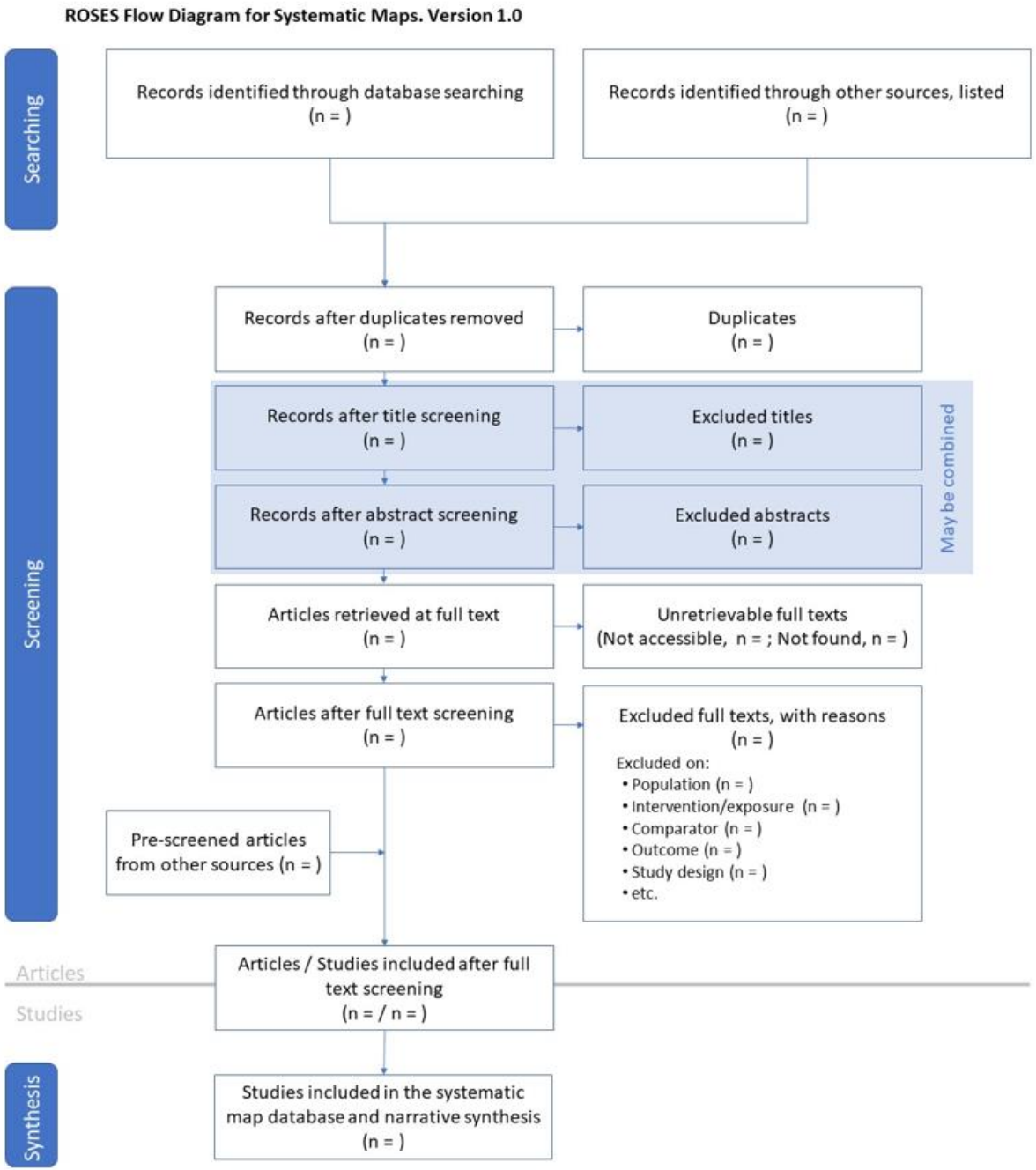
Download the flow diagram (Powerpoint file)

ROSES Flow Diagram for Systematic Maps

Download the flow diagram (Powerpoint file)

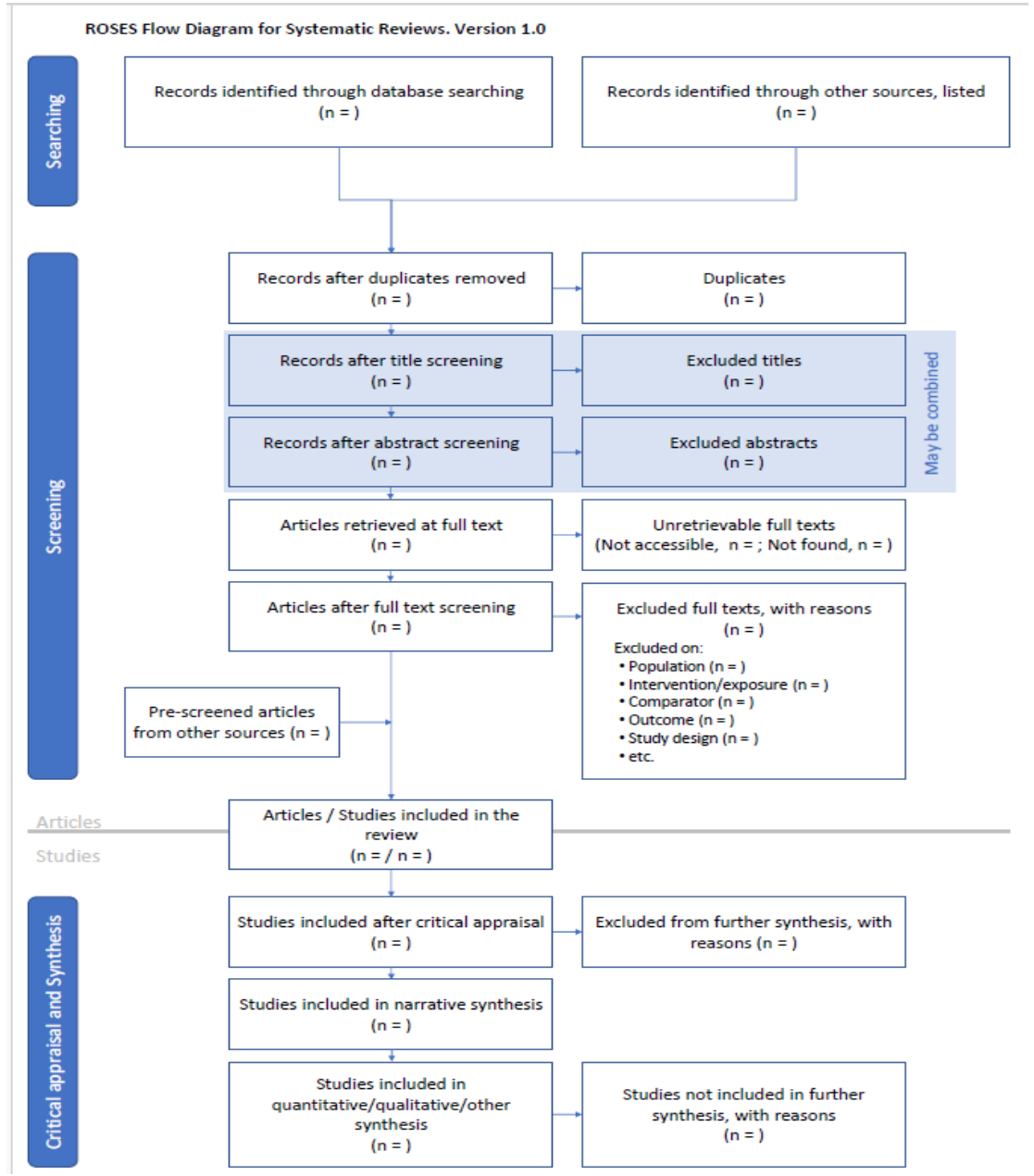
Templates

For Systematic maps

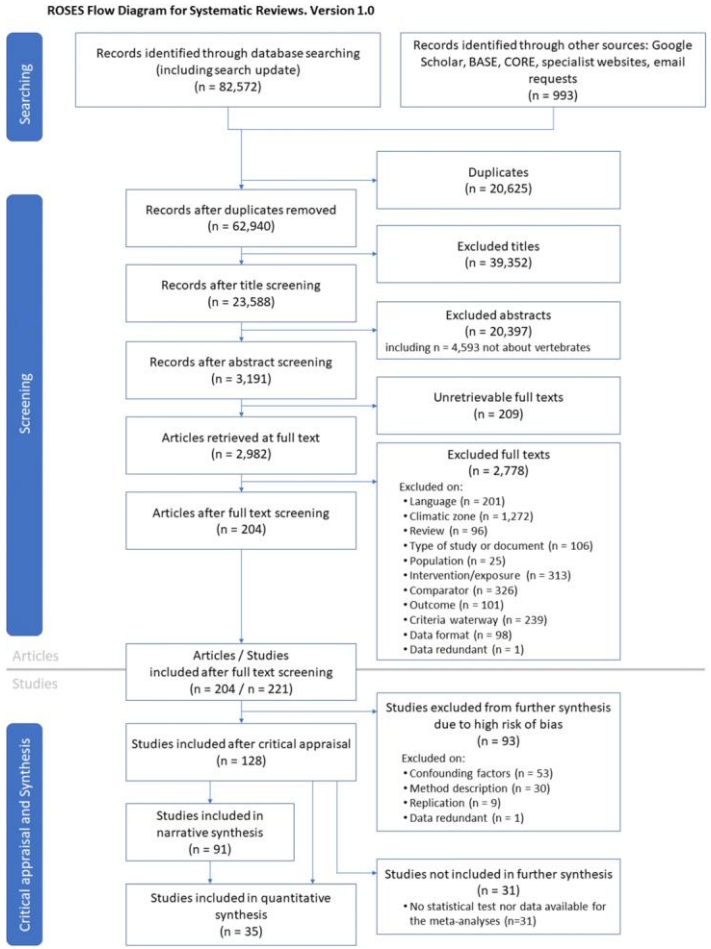


Templates

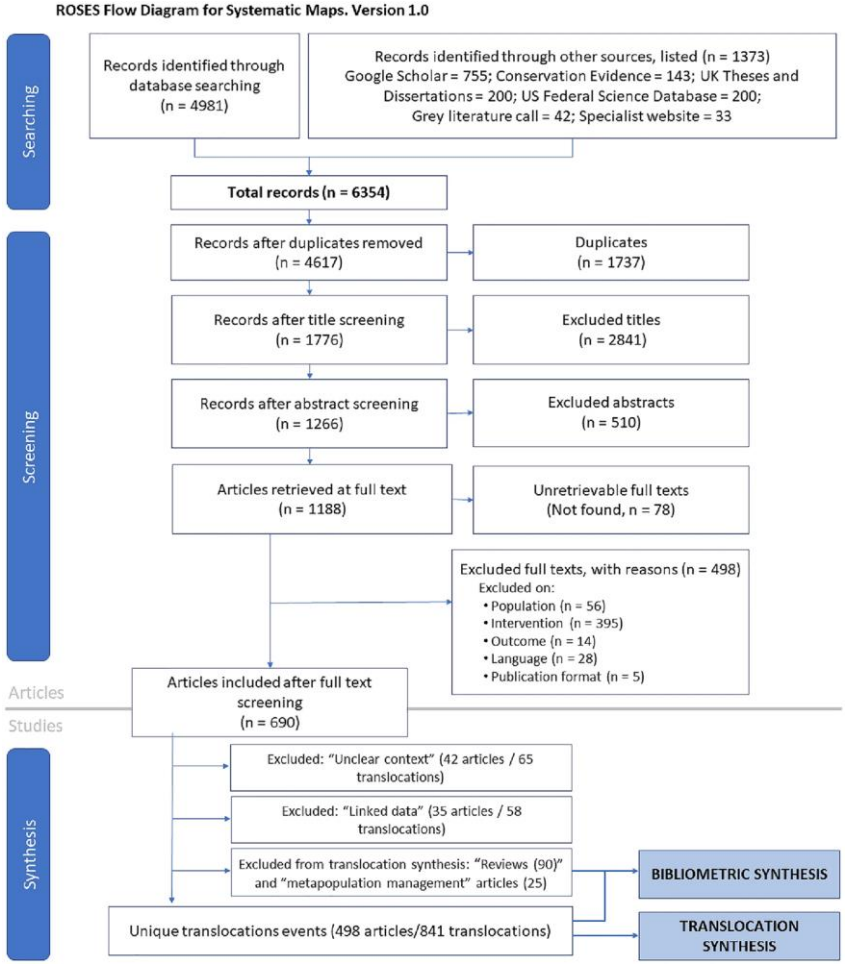
for Systematic Reviews



Example ROSES Flows



Source: Ouédraogo et al. 2020. *Environ Evid*



Source: Langridge et al. 2021. *Environ Evid*

Ensuring transparency and traceability

biblio internal id	biblio authors	biblio title	biblio container	biblio abstract	biblio year	biblio doi	biblio woskey	duplicate check	screen title decision	Articles qualified for abstract screening	screen abstract decision	pdf retrieved	file name	submitted to full text screening	screen fulltext language	screen fulltext pico	screen fulltext pico justif	screen fulltext decision	screen fulltext decision justif
15	Beschta, RL;	Can large carnivores change stream	ECOHYDROLO	Large carnivo	2019	10.1002/eco.	WOS:000454	No: retained for screening	Y	Yes: retained on title	Y	YES	Beschta_et_a	Yes: retained on abstract	en	N	o	N	pico_exclusion
19	Bakker, ES; S	Trophic rewilding: impact on ecosys	PHILOSOPHIC	no_abstract	2018	10.1098/rstb.	WOS:000449	No: retained for screening	Y	Yes: retained on title	No-Abstract	YES	Bakker_and_	Yes: retained by default - no	en	N	i	N	pico_exclusion
27	Walker, LE; M	Population responses of common ra	ECOLOGY AN	Top predator	2018	10.1002/ece3	WOS:000451	No: retained for screening	Y	Yes: retained on title	Y	YES	Walker_et_al	Yes: retained on abstract	en	N	i	N	pico_exclusion
31	Kuiper, TR; D	Demography and social dynamics of	JOURNAL OF	1. Given thei	2018	10.1111/1365	WOS:000447	No: retained for screening	Y	Yes: retained on title	Y	YES	Kuiper_et_al	Yes: retained on abstract	en	Y	NA	Y	NA
32	Buk, KG; van	Conservation of severely fragmente	BIODIVERSITY	We documen	2018	10.1007/s105	WOS:000445	No: retained for screening	Y	Yes: retained on title	Y	YES	Buk_et_al_2(	Yes: retained on abstract	en	Y	NA	Y	NA
34	Boyce, MS	Wolves for Yellowstone: dynamics in	JOURNAL OF	The reintrod	2018	10.1093/jmar	WOS:000448	No: retained for screening	Y	Yes: retained on title	Y	YES	Boyce_2018	Yes: retained on abstract	en	N	i	N	pico_exclusion
39	Sagua, MI; Fi	Inferring the origin and genetic dive	MAMMAL RE	The Eurasian	2018	10.1007/s133	WOS:000444	No: retained for screening	Y	Yes: retained on title	Y	YES	Sagua_et_al	Yes: retained on abstract	en	N	i	N	pico_exclusion
47	Watkins, GE;	Success criteria not met, but valuab	WILDLIFE RES	Context. Fau	2018	10.1071/WR1	WOS:000444	No: retained for screening	Y	Yes: retained on title	Y	YES	Watkins_et_	Yes: retained on abstract	en	Y	NA	Y	NA
50	Painter, LE; B	Aspen recruitment in the Yellowstor	ECOSPHERE	Quaking aspe	2018	10.1002/ecs2	WOS:000448	No: retained for screening	Y	Yes: retained on title	Y	YES	Painter_et_a	Yes: retained on abstract	en	N	o	N	pico_exclusion
51	Zheng, LM; W	Habitat Evaluation for Reintroduced	PAKISTAN JO	Crested ibis (	2018	10.17582/jou	WOS:000447	No: retained for screening	Y	Yes: retained on title	Y	YES	Zheng_et_al	Yes: retained on abstract	en	Y	NA	Y	NA
53	Muller, LI; Mi	Genetic structure in Elk persists afte	JOURNAL OF	Elk (Cervus c	2018	10.1002/jwm	WOS:000439	No: retained for screening	Y	Yes: retained on title	Y	YES	Muller_et_al	Yes: retained on abstract	en	Y	NA	Y	NA
55	Heurich, M; S	Illegal hunting as a major driver of t	BIOLOGICAL	Large carnivo	2018	10.1016/j.bio	WOS:000439	No: retained for screening	Y	Yes: retained on title	Y	YES	Heurich_et_a	Yes: retained on abstract	en	Y	NA	Y	NA
63	Kenup, CF; Se	Walking on their own legs: unassiste	ORYX	Reintroductio	2018	10.1017/S003	WOS:000450	No: retained for screening	Y	Yes: retained on title	Y	YES	Kenup_et_al	Yes: retained on abstract	en	Y	NA	Y	NA
65	Vasile, M	The Vulnerable Bison: Practices and	CONSERVATI	This paper di	2018	10.4103/cs.cs	WOS:000445	No: retained for screening	Y	Yes: retained on title	Y	YES	Vasile_2018	Yes: retained on abstract	en	N	p	N	pico_exclusion
67	Elbroch, LM;	Multiple anthropogenic intervention	ECOLOGY AN	Humans are	2018	10.1002/ece3	WOS:000440	No: retained for screening	Y	Yes: retained on title	Y	YES	Elbroch_et_a	Yes: retained on abstract	en	N	i	N	pico_exclusion
68	Miranda, EB	Reintroducing apex predators: the p	ROYAL SOCIE	no_abstract	2018	10.1098/rsos	WOS:000440	No: retained for screening	Y	Yes: retained on title	No-Abstract	YES	Miranda_201	Yes: retained by default - no	en	N	p	N	pico_exclusion
69	Smith, TN; Ke	Diet Composition and Selection of R	AMERICAN M	Since elk [Cer	2018	10.1674/0003	WOS:000438	No: retained for screening	Y	Yes: retained on title	Y	YES	Smith_et_al	Yes: retained on abstract	en	Y	NA	Y	NA
71	Rodriguez-Ro	The restoration of the endangered S	PEERJ	The translocz	2018	10.7717/peer	WOS:000435	No: retained for screening	Y	Yes: retained on title	Y	YES	Rodriguez-Ro	Yes: retained on abstract	en	Y	NA	Y	NA
73	Cortez, MV; M	Effect of antipredator training on sp	ACTA ORNITH	The Greater	2018	10.3161/0001	WOS:000442	No: retained for screening	Y	Yes: retained on title	Y	YES	Cortez_et_al	Yes: retained on abstract	en	Y	NA	Y	NA
75	Renan, S; Spe	Fission-fusion social structure of a re	BIOLOGICAL	In a reintrod	2018	10.1016/j.bio	WOS:000434	No: retained for screening	Y	Yes: retained on title	Y	YES	Renan_et_al	Yes: retained on abstract	en	Y	NA	Y	NA
77	Woodruff, SP	Characteristics of Winter Wolf Kill Si	JOURNAL OF	Understandir	2018	10.3996/032(	WOS:000434	No: retained for screening	Y	Yes: retained on title	Y	YES	Woodruff_et	Yes: retained on abstract	en	Y	NA	Y	NA
87	Hale, SL; Kop	Ecosystem-level effects of keystone	RESTORATIO	The keystone	2018	10.1111/rec.1	WOS:000434	No: retained for screening	Y	Yes: retained on title	Y	YES	Hale_and_Ko	Yes: retained on abstract	en	N	i	N	pico_exclusion
96	Makin, DF; Cl	Changes in feeding behavior and pa	JOURNAL OF	Top-order ca	2018	10.1093/jmar	WOS:000429	No: retained for screening	Y	Yes: retained on title	Y	YES	Makin_et_al	Yes: retained on abstract	en	N	i	N	pico_exclusion
99	Codron, D; Ri	Meso-carnivore niche expansion in t	AFRICAN JOU	Apex predato	2018	10.3957/056.	WOS:000432	No: retained for screening	Y	Yes: retained on title	Y	YES	Codron_et_a	Yes: retained on abstract	en	N	o	N	pico_exclusion
101	Abeli, T; Cau	A gleam of hope for the critically en	AQUATIC CO	Results of the	2018	10.1002/aqc.	WOS:000429	No: retained for screening	Y	Yes: retained on title	Y	YES	Abeli_et_al_	Yes: retained on abstract	en	Y	NA	Y	NA
109	Jolly, CJ; Kelly	Out of the frying pan: Reintroductio	AUSTRAL ECO	Invasive spec	2018	10.1111/aec.	WOS:000428	No: retained for screening	Y	Yes: retained on title	Y	YES	Jolly_et_al_2	Yes: retained on abstract	en	Y	NA	Y	NA
110	Beschta, RL;	Trophic cascades at multiple spatial	FOREST ECOL	Throughout r	2018	10.1016/j.for	WOS:000427	No: retained for screening	Y	Yes: retained on title	Y	YES	Beschta_et_a	Yes: retained on abstract	en	N	o	N	pico_exclusion
112	Zarrouk, A; Ri	Reintroduction of the highly endang	ESTUARINE C	Patella ferrug	2018	10.1016/j.ecs	WOS:000426	No: retained for screening	Y	Yes: retained on title	Y	YES	Zarrouk_et_a	Yes: retained on abstract	en	Y	NA	Y	NA
113	Stover, HJ; Na	Transplanting Following Non-Native	ECOLOGICAL	Disturbed are	2018	10.3368/er.3(	WOS:000456	No: retained for screening	Y	Yes: retained on title	Y	YES	Stover_et_al	Yes: retained on abstract	en	Y	NA	Y	NA
116	Scrafford, MA	Beaver Habitat Selection for 24 Yr Si	RANGELAND	Beavers (Cast	2018	10.1016/j.ran	WOS:000426	No: retained for screening	Y	Yes: retained on title	Y	YES	Scrafford_et	Yes: retained on abstract	en	Y	NA	Y	NA
128	Valko, O; Tot	Cultural heritage and biodiversity co	NATURE CON	Linking the co	2018	10.3897/natu	WOS:000419	No: retained for screening	Y	Yes: retained on title	Y	YES	Valko_et_al_	Yes: retained on abstract	en	Y	NA	Y	NA
131	Bakka, SV; Ki	The story of the creation and monit	ALL-RUSSIAN	The article su	2018	10.1088/1755	WOS:000454	No: retained for screening	Y	Yes: retained on title	Y	YES	Bakka_et_al_	Yes: retained on abstract	en	Y	NA	Y	NA
143	Vogel, JT; Sor	The foraging ecology of reintroduce	WILDLIFE BIO	Restoring lar	2018	10.2981/wlb.	WOS:000439	No: retained for screening	Y	Yes: retained on title	Y	YES	Vogel_et_al_	Yes: retained on abstract	en	Y	NA	Y	NA
146	Robinson, NM	Bandicoots return to Booderee: init	WILDLIFE RES	Context. Rein	2018	10.1071/WR1	WOS:000431	No: retained for screening	Y	Yes: retained on title	Y	YES	Robinson_et	Yes: retained on abstract	en	Y	NA	Y	NA
147	Ismaili, B; Dic	Getting the dietary knowledge to re	WILDLIFE BIO	Atlas deer Ce	2018	10.2981/wlb.	WOS:000430	No: retained for screening	Y	Yes: retained on title	Y	YES	Ismaili_et_al	Yes: retained on abstract	en	Y	NA	Y	NA
153	Leung, FTY; Y	MECHANICAL PULL-OUT CAPACITY A	JOURNAL OF	Ecological rel	2018	10.26525/jtfs	WOS:000422	No: retained for screening	Y	Yes: retained on title	Y	YES	Leung_et_al	Yes: retained on abstract	en	N	i	N	pico_exclusion
154	Brzosko, E; Je	Rare plant translocation between m	RESTORATIO	We carried o	2018	10.1111/rec.1	WOS:000419	No: retained for screening	Y	Yes: retained on title	Y	YES	Brzosko_et_a	Yes: retained on abstract	en	Y	NA	Y	NA
159	Wu, J; Zhu, JC	Breeding Ecology of a Reintroduced	PAKISTAN JO	The crested i	2017	10.17582/jou	WOS:000424	No: retained for screening	Y	Yes: retained on title	Y	YES	Wu_et_al_2(	Yes: retained on abstract	en	Y	NA	Y	NA
160	Kovacic, I; Pa	Adaptation of cultured mussel Mytil	ACTA ADRIAT	The ability of	2017	no_DOI	WOS:000423	No: retained for screening	Y	Yes: retained on title	Y	YES	Kovacic_et_a	Yes: retained on abstract	en	N	i	N	pico_exclusion
161	Cremona, T;	High mortality and small population	ANIMAL CON	Northern Aus	2017	10.1111/acv.	WOS:000423	No: retained for screening	Y	Yes: retained on title	Y	YES	Cremona_et	Yes: retained on abstract	en	Y	NA	Y	NA
167	Cremona, T;	Avoiding the last supper: parentage	CONSERVATI	Invasive spec	2017	10.1007/s105	WOS:000414	No: retained for screening	Y	Yes: retained on title	Y	YES	Cremona_et	Yes: retained on abstract	en	Y	NA	Y	NA
171	Sarkar, MS; K	Assessment of fine-scale resource se	PEERJ	Background.	2017	10.7717/peer	WOS:000414	No: retained for screening	Y	Yes: retained on title	Y	YES	Sarkar_et_al	Yes: retained on abstract	en	Y	NA	Y	NA
176	McClure, CJW	Successful enhancement of Ridgway	CONDOR	Survival and	2017	10.1650/CON	WOS:000415	No: retained for screening	Y	Yes: retained on title	Y	YES	McClure_et_	Yes: retained on abstract	en	Y	NA	Y	NA
180	Benjamin-Fin	Conservation implications of wildlife	GLOBAL ECO	Wildlife trans	2017	10.1016/j.gec	WOS:000426	No: retained for screening	Y	Yes: retained on title	Y	YES	Benjamin-Fin	Yes: retained on abstract	en	N	p	N	pico_exclusion
183	Zamboni, T; E	A review of a multispecies reintrodu	PERSPECTIVE	Ibera Nature	2017	10.1016/j.pec	WOS:000418	No: retained for screening	Y	Yes: retained on title	Y	YES	Zamboni_et	Yes: retained on abstract	en	Y	NA	Y	NA
184	Fernandez, F	Rewilding the Atlantic Forest: Resto	PERSPECTIVE	The loss or re	2017	10.1016/j.pec	WOS:000418	No: retained for screening	Y	Yes: retained on title	Y	YES	Fernandez_e	Yes: retained on abstract	en	Y	NA	Y	NA

Source: Langridge et al. 2021. *Environ Evid*

Ensuring transparency and traceability

biblio internal id	biblio authors	biblio title	biblio container	biblio abstract	biblio year	biblio doi	biblio woskey	duplicate check	screen title decision	Articles qualified for abstract screening	screen abstract decision	pdf retrieved	file name	submitted to full text screening	screen fulltext language	screen fulltext pico	screen fulltext pico justif	screen fulltext decision	screen fulltext decision justif
15	Beschta, RL;	Can large carnivores change stream	ECOHYDROLO	Large carnivo	2019	10.1002/eco.	WOS:000454	No: retained for screening	Y	Yes: retained on title	Y	YES	Beschta_et_a	Yes: retained on abstract	en	N	o	N	pico_exclusion
19	Bakker, ES; S	Trophic rewilding: impact on ecosys	PHILOSOPHIC	no_abstract	2018	10.1098/rstb.	WOS:000449	No: retained for screening	Y	Yes: retained on title	No-Abstract	YES	Bakker_and_	Yes: retained by default - no	en	N	i	N	pico_exclusion
27	Walker, LE; M	Population responses of common ra	ECOLOGY AN	Top predator	2018	10.1002/ece3	WOS:000451	No: retained for screening	Y	Yes: retained on title	Y	YES	Walker_et_al	Yes: retained on abstract	en	N	i	N	pico_exclusion
31	Kuiper, TR; D	Demography and social dynamics of	JOURNAL OF	1. Given thei	2018	10.1111/1365	WOS:000447	No: retained for screening	Y	Yes: retained on title	Y	YES	Kuiper_et_al	Yes: retained on abstract	en	Y	NA	Y	NA
32	Buk, KG; van	Conservation of severely fragmente	BIODIVERSITY	We documen	2018	10.1007/s105	WOS:000445	No: retained for screening	Y	Yes: retained on title	Y	YES	Buk_et_al_2(	Yes: retained on abstract	en	Y	NA	Y	NA
34	Boyce, MS	Wolves for Yellowstone: dynamics in	JOURNAL OF	The reintrod	2018	10.1093/jmar	WOS:000448	No: retained for screening	Y	Yes: retained on title	Y	YES	Boyce_2018	Yes: retained on abstract	en	N	i	N	pico_exclusion
39	Sagua, MI; Fi	Inferring the origin and genetic dive	MAMMAL RE	The Eurasian	2018	10.1007/s133	WOS:000444	No: retained for screening	Y	Yes: retained on title	Y	YES	Sagua_et_al	Yes: retained on abstract	en	N	i	N	pico_exclusion
47	Watkins, GE;	Success criteria not met, but valuab	WILDLIFE RES	Context. Fau	2018	10.1071/WR1	WOS:000444	No: retained for screening	Y	Yes: retained on title	Y	YES	Watkins_et_	Yes: retained on abstract	en	Y	NA	Y	NA
50	Painter, LE; B	Aspen recruitment in the Yellowstor	ECOSPHERE	Quaking aspe	2018	10.1002/ecs2	WOS:000448	No: retained for screening	Y	Yes: retained on title	Y	YES	Painter_et_a	Yes: retained on abstract	en	N	o	N	pico_exclusion
51	Zheng, LM; W	Habitat Evaluation for Reintroduced	PAKISTAN JO	Crested ibis (	2018	10.17582/jou	WOS:000447	No: retained for screening	Y	Yes: retained on title	Y	YES	Zheng_et_al	Yes: retained on abstract	en	Y	NA	Y	NA
53	Muller, LI; M	Genetic structure in Elk persists afte	JOURNAL OF	Elk (Cervus c	2018	10.1002/jwm	WOS:000439	No: retained for screening	Y	Yes: retained on title	Y	YES	Muller_et_al	Yes: retained on abstract	en	Y	NA	Y	NA
55	Heurich, M; S	Illegal hunting as a major driver of t	BIOLOGICAL (	Large carnivo	2018	10.1016/j.bio	WOS:000439	No: retained for screening	Y	Yes: retained on title	Y	YES	Heurich_et_a	Yes: retained on abstract	en	Y	NA	Y	NA
63	Kenup, CF; Se	Walking on their own legs: unassiste	ORYX	Reintroductic	2018	10.1017/S003	WOS:000450	No: retained for screening	Y	Yes: retained on title	Y	YES	Kenup_et_al	Yes: retained on abstract	en	Y	NA	Y	NA
65	Vasile, M	The Vulnerable Bison: Practices and	CONSERVATI	This paper di	2018	10.4103/cs.cs	WOS:000445	No: retained for screening	Y	Yes: retained on title	Y	YES	Vasile_2018	Yes: retained on abstract	en	N	p	N	pico_exclusion
67	Elbroch, LM;	Multiple anthropogenic interventio	ECOLOGY AN	Humans are	2018	10.1002/ece3	WOS:000440	No: retained for screening	Y	Yes: retained on title	Y	YES	Elbroch_et_a	Yes: retained on abstract	en	N	i	N	pico_exclusion
									ing	Y	Yes: retained on title	No-Abstract	YES	Miranda_201	Yes: retained by default - no	en	N	p	N	pico_exclusion
									ing	Y	Yes: retained on title	Y	YES	Smith_et_al	Yes: retained on abstract	en	Y	NA	Y	NA
									ing	Y	Yes: retained on title	Y	YES	Rodriguez-Rc	Yes: retained on abstract	en	Y	NA	Y	NA
									ing	Y	Yes: retained on title	Y	YES	Cortez_et_al	Yes: retained on abstract	en	Y	NA	Y	NA
									ing	Y	Yes: retained on title	Y	YES	Renan_et_al	Yes: retained on abstract	en	Y	NA	Y	NA
									ing	Y	Yes: retained on title	Y	YES	Woodruff_et	Yes: retained on abstract	en	Y	NA	Y	NA
									ing	Y	Yes: retained on title	Y	YES	Hale_and_Ko	Yes: retained on abstract	en	N	i	N	pico_exclusion
									ing	Y	Yes: retained on title	Y	YES	Makin_et_al	Yes: retained on abstract	en	N	i	N	pico_exclusion
									ing	Y	Yes: retained on title	Y	YES	Codron_et_a	Yes: retained on abstract	en	N	o	N	pico_exclusion
101	Abeli, T; Cau	A gleam of hope for the critically en	AQUATIC CO	Results of the	2018	10.1002/aqc.	WOS:000429	No: retained for screening	Y	Yes: retained on title	Y	YES	Abeli_et_al_	Yes: retained on abstract	en	Y	NA	Y	NA
109	Jolly, CJ; Kelly	Out of the frying pan: Reintroductio	AUSTRAL ECO	Invasive spec	2018	10.1111/aec.	WOS:000428	No: retained for screening	Y	Yes: retained on title	Y	YES	Jolly_et_al_2	Yes: retained on abstract	en	Y	NA	Y	NA
110	Beschta, RL;	Trophic cascades at multiple spatial	FOREST ECOL	Throughout r	2018	10.1016/j.for	WOS:000427	No: retained for screening	Y	Yes: retained on title	Y	YES	Beschta_et_a	Yes: retained on abstract	en	N	o	N	pico_exclusion
112	Zarrouk, A; R	Reintroduction of the highly endang	ESTUARINE C	Patella ferrug	2018	10.1016/j.ecs	WOS:000426	No: retained for screening	Y	Yes: retained on title	Y	YES	Zarrouk_et_a	Yes: retained on abstract	en	Y	NA	Y	NA
113	Stover, HJ; Na	Transplanting Following Non-Native	ECOLOGICAL	Disturbed are	2018	10.3368/er.3(	WOS:000456	No: retained for screening	Y	Yes: retained on title	Y	YES	Stover_et_al	Yes: retained on abstract	en	Y	NA	Y	NA
116	Scrafford, MA	Beaver Habitat Selection for 24 Yr Si	RANGELAND	Beavers (Cast	2018	10.1016/j.ran	WOS:000426	No: retained for screening	Y	Yes: retained on title	Y	YES	Scrafford_et	Yes: retained on abstract	en	Y	NA	Y	NA
128	Valko, O; Tot	Cultural heritage and biodiversity co	NATURE CON	Linking the co	2018	10.3897/natu	WOS:000419	No: retained for screening	Y	Yes: retained on title	Y	YES	Valko_et_al_	Yes: retained on abstract	en	Y	NA	Y	NA
131	Bakka, SV; Ki	The story of the creation and monit	ALL-RUSSIAN	The article su	2018	10.1088/1755	WOS:000454	No: retained for screening	Y	Yes: retained on title	Y	YES	Bakka_et_al_	Yes: retained on abstract	en	Y	NA	Y	NA
143	Vogel, JT; Sor	The foraging ecology of reintroduce	WILDLIFE BIO	Restoring lar	2018	10.2981/wlb.	WOS:000439	No: retained for screening	Y	Yes: retained on title	Y	YES	Vogel_et_al_	Yes: retained on abstract	en	Y	NA	Y	NA
146	Robinson, NM	Bandicoots return to Booderee: init	WILDLIFE RES	Context. Rein	2018	10.1071/WR1	WOS:000431	No: retained for screening	Y	Yes: retained on title	Y	YES	Robinson_et	Yes: retained on abstract	en	Y	NA	Y	NA
147	Ismaili, B; Dic	Getting the dietary knowledge to re	WILDLIFE BIO	Atlas deer Ce	2018	10.2981/wlb.	WOS:000430	No: retained for screening	Y	Yes: retained on title	Y	YES	Ismaili_et_al	Yes: retained on abstract	en	Y	NA	Y	NA
153	Leung, FTY; Y	MECHANICAL PULL-OUT CAPACITY A	JOURNAL OF	Ecological rel	2018	10.26525/jtfs	WOS:000422	No: retained for screening	Y	Yes: retained on title	Y	YES	Leung_et_al	Yes: retained on abstract	en	N	i	N	pico_exclusion
154	Brzosko, E; Je	Rare plant translocation between m	RESTORATIOI	We carried o	2018	10.1111/rec.1	WOS:000419	No: retained for screening	Y	Yes: retained on title	Y	YES	Brzosko_et_a	Yes: retained on abstract	en	Y	NA	Y	NA
159	Wu, J; Zhu, J(	Breeding Ecology of a Reintroduced	PAKISTAN JO	The crested i	2017	10.17582/jou	WOS:000424	No: retained for screening	Y	Yes: retained on title	Y	YES	Wu_et_al_2(	Yes: retained on abstract	en	Y	NA	Y	NA
160	Kovacic, I; Par	Adaptation of cultured mussel Mytil	ACTA ADRIAT	The ability of	2017	no_DOI	WOS:000423	No: retained for screening	Y	Yes: retained on title	Y	YES	Kovacic_et_a	Yes: retained on abstract	en	N	i	N	pico_exclusion
161	Cremona, T; (	High mortality and small population	ANIMAL CON	Northern Aus	2017	10.1111/acv.	WOS:000423	No: retained for screening	Y	Yes: retained on title	Y	YES	Cremona_et	Yes: retained on abstract	en	Y	NA	Y	NA
167	Cremona, T; (	Avoiding the last supper: parentage	CONSERVATI	Invasive spec	2017	10.1007/s105	WOS:000414	No: retained for screening	Y	Yes: retained on title	Y	YES	Cremona_et	Yes: retained on abstract	en	Y	NA	Y	NA
171	Sarkar, MS; K	Assessment of fine-scale resource se	PEERJ	Background.	2017	10.7717/peer	WOS:000414	No: retained for screening	Y	Yes: retained on title	Y	YES	Sarkar_et_al	Yes: retained on abstract	en	Y	NA	Y	NA
176	McClure, CJW	Successful enhancement of Ridgway	CONDOR	Survival and	2017	10.1650/CON	WOS:000415	No: retained for screening	Y	Yes: retained on title	Y	YES	McClure_et_	Yes: retained on abstract	en	Y	NA	Y	NA
180	Benjamin-Fin	Conservation implications of wildlif	GLOBAL ECO	Wildlife trans	2017	10.1016/j.gec	WOS:000426	No: retained for screening	Y	Yes: retained on title	Y	YES	Benjamin-Fin	Yes: retained on abstract	en	N	p	N	pico_exclusion
183	Zamboni, T; E	A review of a multispecies reintrodu	PERSPECTIVE	Ibera Nature	2017	10.1016/j.pec	WOS:000418	No: retained for screening	Y	Yes: retained on title	Y	YES	Zamboni_et	Yes: retained on abstract	en	Y	NA	Y	NA
184	Fernandez, F	Rewilding the Atlantic Forest: Resto	PERSPECTIVE	The loss or re	2017	10.1016/j.pec	WOS:000418	No: retained for screening	Y	Yes: retained on title	Y	YES	Fernandez_e	Yes: retained on abstract	en	Y	NA	Y	NA

Minimum requirement
: provide full text exclusion decisions !

Source: Langridge et al. 2021. *Environ Evid*

ROSES

forms

ROSES **forms**

Why is it important?

- **Facilitates manuscript writing for authors**: *helps ensure that the all correct information has been included with the appropriate level of detail adhering to* Collaboration for Environmental Evidence guidelines

- **Supports the writing process**: avoid manuscript revisions or rejections

- **Serves as a "checklist"**: really useful if publishing in Environmental Evidence Journal (EEJ)

- **Serves as a guarantee of transparency and rigor**.

ROSES **forms**

Comment | Published: 30 May 2018

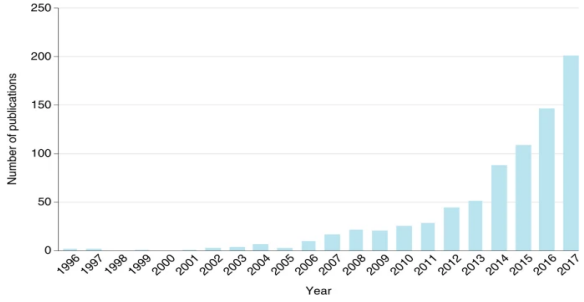
The role of reporting standards in producing robust literature reviews

Neal Robert Haddaway & Biljana Macura

Nature Climate Change **8**, 444–447 (2018) | Cite this article

2452 Accesses | **79** Citations | **63** Altmetric | Metrics

Fig. 1: Increasing interest in systematic reviews over recent years.

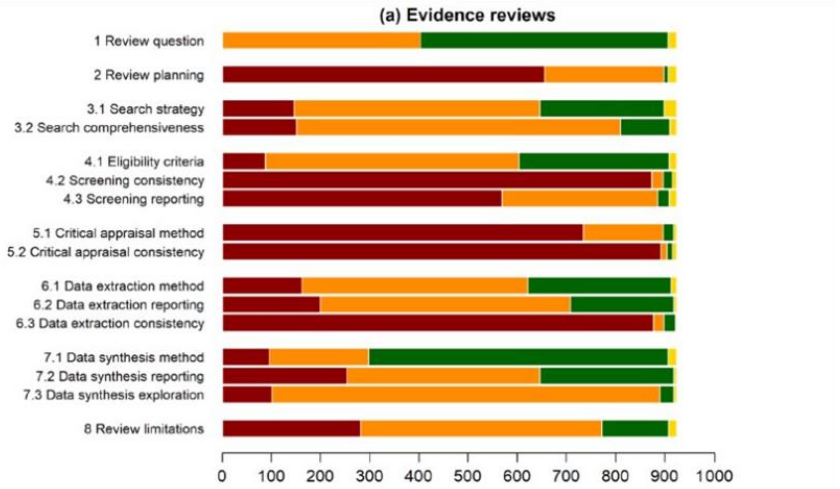


Full size image >

The number of publications returned from a search of the Web of Science Core Collections for 'systematic review' in topic words within the 'Environmental science' subject category on 14 April 2018 is shown, divided into annual bins.

Source: Haddaway & Macura. 2018. *Nature climate change.*

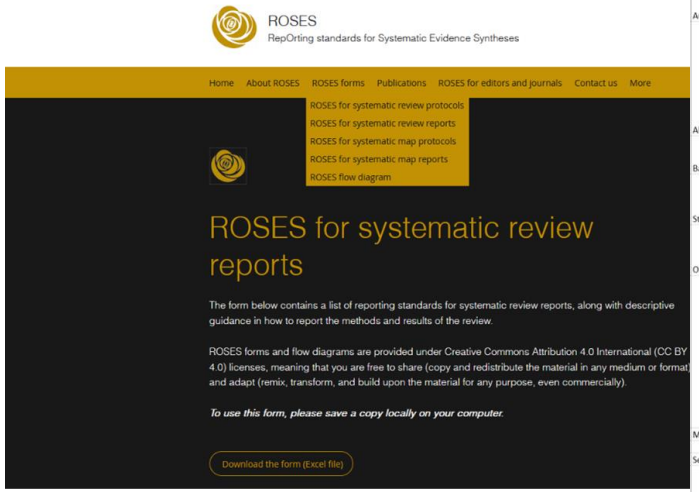
Increasing interest but poor consideration for reporting and transparency.



Source: Pullin *et al.* 2022. *Environ Evid*

ROSES forms for **systematic maps**

Essentially a check list



Type of review	Type of review	Select one of the following types of review: systematic map, systematic map update, systematic map amendment	See CEE Guidance on systematic mapping [1], and on amendments and updates [2]	Meta-data	systematic map	
Authors' contacts	Authors' contacts	The full names, institutional addresses and email addresses for all authors must be provided.		Checklist	Yes	
Abstract	Structured summary	The abstract of the manuscript must not exceed 500 words and must be structured into separate sections: Background, the context and purpose of the review, including the review question; Methods, how the review was performed (specifically mention search strategy, inclusion criteria, critical appraisal (optional), meta-data extraction and coding, and narrative synthesis); Results, the main findings, including results of search and assessment of evidence base; Conclusions, brief summary and potential implications for policy/management and research.		Checklist	Yes	
Background	Background	Describe the rationale for the review in the context of what is already known. Reviews must indicate why this study was necessary and what it aims to contribute to the field.	A theory of change and/or conceptual model should be presented that links the intervention or exposure to the outcome.	Checklist	Yes	
Stakeholder engagement	Stakeholder engagement	The actual role of stakeholders throughout the review process (e.g. in the formulation of the question) must be described and explained (using a broad definition of 'stakeholder', including e.g. researchers, funders and other decision-makers; see [3])		Checklist	Yes	
Objective of the review	Objective	Describe the primary question and secondary questions (when applicable).	The primary question is the main question of the review. The secondary questions are usually linked to sources of heterogeneity (effect modifiers).	Checklist	Yes	
	Definition of the question components	Provide reference to the question key elements, e.g. population(s), intervention(s)/exposure(s), comparator(s), and outcome(s).	For other question types see [4,5]	Meta-data	Population (P): All species (fauna, flora fungi) of wild or captive source. Intervention (I): All conservation-based translocations (reintroductions, supplementations (reinforcements) and introductions (assisted migrations)). Comparator (C): Studies will not be required *stricto sensu* to have a comparator. Outcomes (O): All relevant outcomes related to the translocated population e.g. survival, demography, space use, behavior, reproduction.	Refer to Table 1 of mauscrit for full breakdown
Methods	Protocol	Provide citation, DOI or open-access link to published protocol.	The protocol should be peer-reviewed and	Meta-data	https://doi.org/10.1186/s13750-020-00199-4	
	Deviations from protocol	Describe any ways in which the final methods of the review deviate		Checklist	Yes	
Searches	Search strategy	Detail the search strategy used, including: database names accessed,		Checklist	Yes	
	Search string	Provide Boolean-style full search string and state the platform for which the string is formatted (e.g. Web of Science format)		Meta-data	TS = ("protected area$" OR "natur* reserve$" OR "national park$" OR "regional park$" OR "national reserve$" OR "biological reserve$" OR "regional reserve$") AND TS = ("assisted colonization" OR "assisted population migration" OR "assisted migration" OR "assisted gene flow" OR "managed relocation$" OR transloc* OR reintroduc* OR reinforc* OR "assisted range expansion$" OR "assisted long-distance migration$").	
	Languages - bibliographic	List languages used in bibliographic database searches		Meta-data	English and French	
	Languages – grey literature	List languages used in organisational website searches and web-		Meta-data	English and French	
	Bibliographic databases	Provide the number of bibliographic databases searched		Meta-data	Two	Web of science (core collection) ; Scopus
	Web-based search engines	Provide the number of web-based search engines searched		Meta-data	Four	Google Scholar, Conservation Evidence, UK Theses and Dissertations, US Federal
	Organisational websites	Provide the number of organisational websites searched		Meta-data	Five	**1) US Fish and wildlife service** (https ://www.fws.gov/). **2) Office National de la Chasse et de la Faune Sauvage** (http://www.oncfs.gouv.fr/) **3) IUCN Conservation Planning Specialist Group's document library** (http://www.cpsg.org/document-repository). **4) Association of Zoos & Aquariums** (https://www.aza.org/). Including the European Association (https://www.eaza.net/). **5) Rewilding Europe** (https ://rewil dinge urope .com/).

Source: Langridge et al. 2021. *Environ Evid*

MESSAGES CLÉS

1. Being transparent ensures trust and credibility

2. Reporting screening steps help when « updating » or « upgrading »

3. Roses forms serve as a “checklist”

4. Can facilitate manuscript writing for authors